

- (6.) Slowly pour or draw the solution through the filter into a sterile bottle. Rinse the filter with 10 mL sterile buffer to maximize yield. ☒
- (7.) Label the sterilized solution with the word “sterile-filtered” or “autoclaved”.

» **Storage of prepared buffers and solutions**

- (1.) Label each container clearly and persistently. Labels must be understood by everyone. Include:

- Name and concentration of the solution
- *Optional:* Recipe identifier
- Hazard labels (prominently, as applicable)



Note: This applies to solutions exceeding 100 mL that contain more than 1–10% flammable liquids or strong corrosives, as well as any solution exceeding 10 mL that contains a particularly hazardous substance such as toxic materials, carcinogens, reproductive toxins, or mixtures that pose serious health hazards or hazards to the environment.

- Date (MM/YYYY) of preparation or expiry (as applicable)
 - Your initials
- (2.) Protect light-sensitive solutions such as solutions containing certain antioxidants by wrapping containers in foil or storing in a dark cabinet, fridge, or freezer.
- (3.) *Optional:* For long-term storage (weeks to months) of frozen goods, aliquot into smaller volumes to reduce freeze-thaw cycles. Write the volume aliquoted on the secondary container. ☒

Quality assurance: Aliquot buffers containing labile components such as EDTA-free protease inhibitors into single-use volumes. Aliquots stored refrigerated or frozen should occupy at least half of the container volume. ♦

- (4.) Store at the recommended temperature.

Hint: Most common buffers can be stored at room temperature. Buffers containing serum, antibiotics, or unstable reagents such as DTT should be kept at 4 °C. Avoid repeated freeze-thawing of frozen stocks.

Critical: Do not store stocks in a frost-free freezer! ◀

- (5.) Maintain a digital (or paper) inventory log of stored solutions and aliquoted materials 📄 SOP0036. Record any deviations from the standard recipe, the date of preparation, when the container was opened, and when the batch was discarded.

Hint: Periodically inspect and discard any solutions that appear cloudy, have precipitates, or are past their shelf life.

Troubleshooting

Weighing out solid reagents

In Step 4:

- The balance does not zero in
 - If you are using an analytical balance, close all doors when weighing. These balances really are sensitive and will be affected by ventilation or someone passing by.

In Step 6:

- Hygroscopic substances
 - Tare the container first, weigh quickly, close the container.
- Static substances
 - Use an antistatic brush or ionizing blower if available.
 - Avoid plastic weigh boats.
 - Ground yourself by touching a metal surface before weighing.

Mixing

In Step 3:

- Stir bar loses rhythm
 - Turn off the stirrer, wait until the stir bar settles, and turn the stirrer on very slowly.

In Step 4:

- The reagent does not dissolve readily
 - Some reagents dissolve slowly or incompletely without heat or pH adjustment. Consult the protocol or solubility table. Find a stir plate with a heater and warm the solution to 37–50 °C while stirring. However, all solutions must be cooled to room temperature before you adjust the pH.
 - Use a finer powder, crush lumps with a spatula, or use mortar and pestle to grind.
 - Use a different salt form with higher solubility such as Na⁺ or K⁺ salt if permitted.
- Hydrophobic or amphipathic compounds clump or float
 - Dissolve in a small volume of ethanol, DMSO, or another compatible organic solvent before adding water.
 - Sonicate if appropriate.
 - Use a mild detergent such as Tween™ 20 if necessary and permitted.

Adjusting the pH by titration

In Step 6:

- Readings are erratic
 - Check for temperature fluctuations; allow the solution to equilibrate.
 - Check whether the electrode buffer may need a refill.
 - If the electrode is cracked or broken, the electrode must be replaced.
 - If a silver/silver chloride reference electrode is used, the silver ions can react with proteins and related substances such as Tris and produce an insoluble precipitate in the porous liquid junction of the electrode. Use an electrode with special junction or a calomel reference element.

Sterilizing buffers and solutions

In Step 3:

- Liquid boiled over or caps popped off
 - Reduce fill volume to 80% of container capacity.
 - Use slow exhaust or shorter sterilization time.

In Step 6:

- Filter clogs or flow is very slow
 - Pre-clarify turbid solutions by low-speed centrifugation and decant supernatant.
 - Use a larger-pore pre-filter (0.45 µm) before sterile filtration with 0.2 µm pore size.

Change log

2021-12-20	Benjamin C. Buchmuller	Adaptation as SOP.
2023-11-19	Benjamin C. Buchmuller	Reformatting.
2026-02-25	Benjamin C. Buchmuller	Language improvement in formula section; add water quality and autoclave validation note.

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